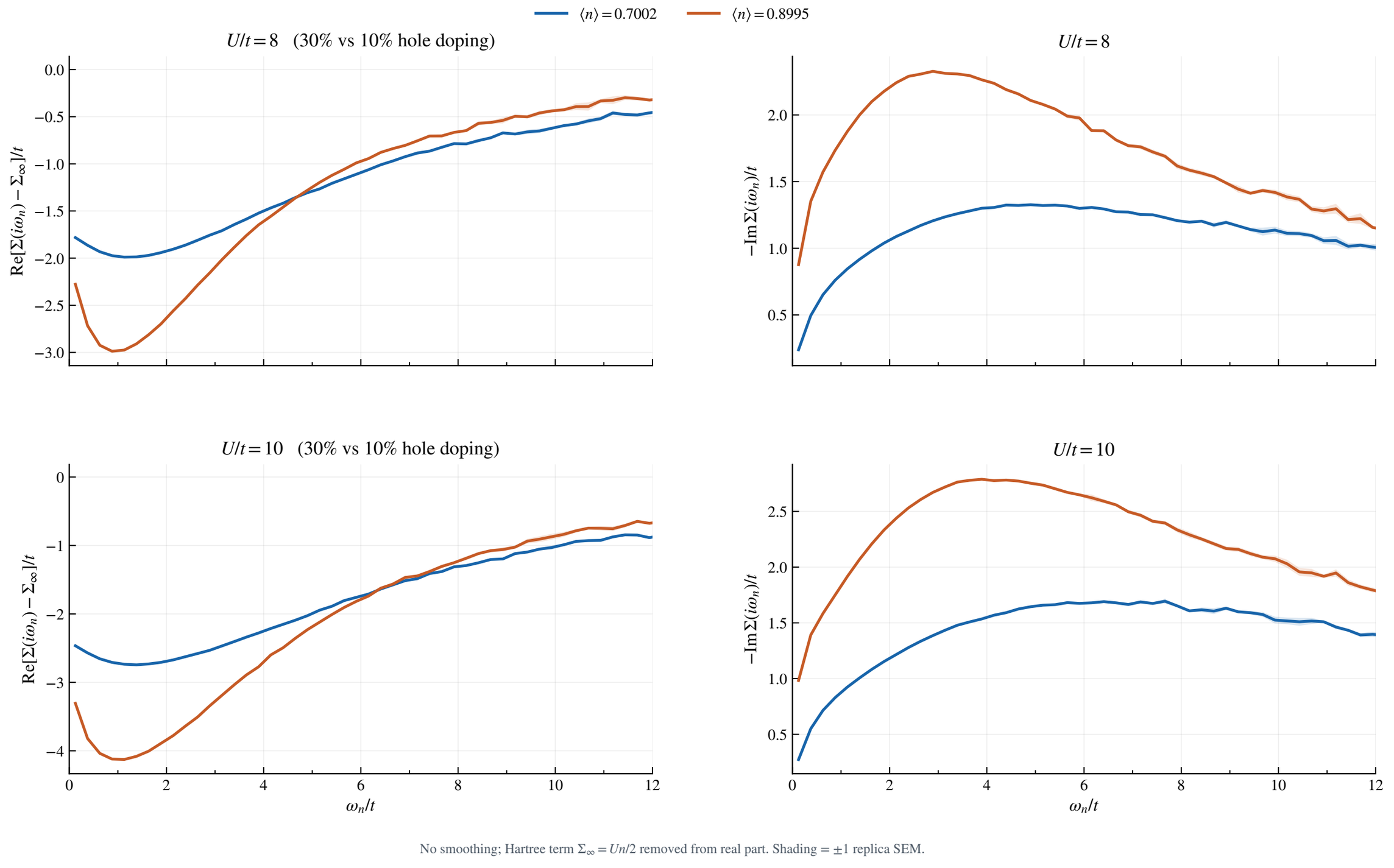
Self-energy: effect of doping at fixed interaction

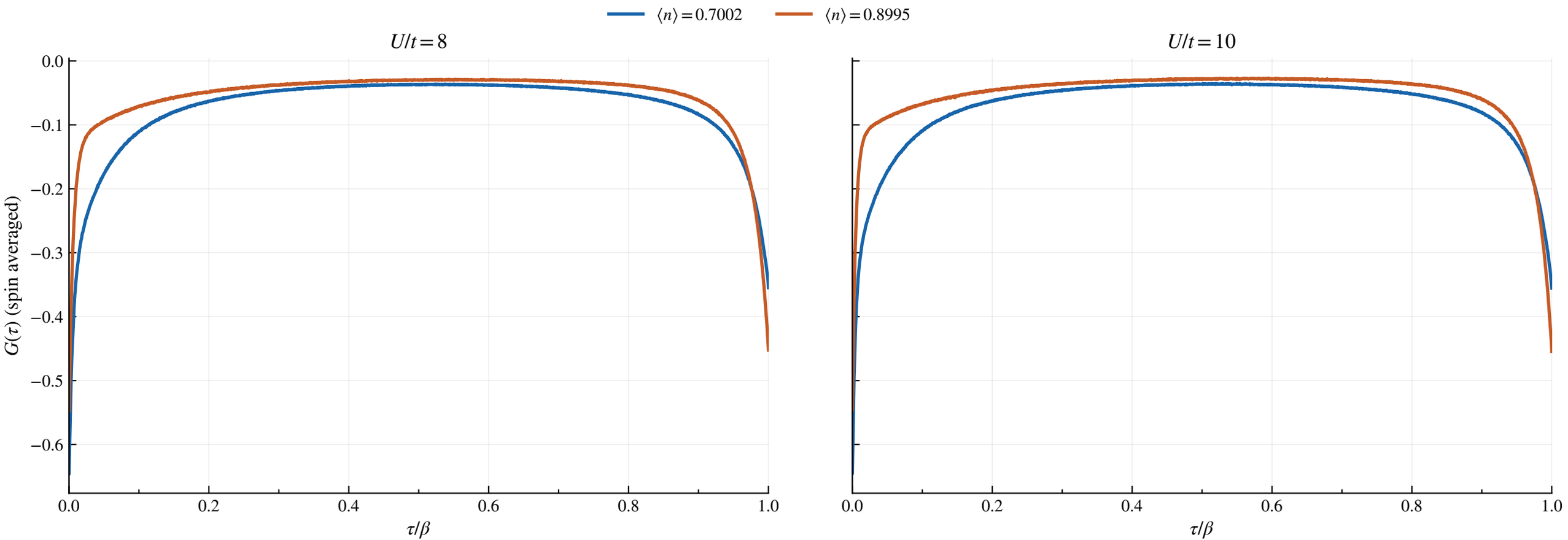
Paramagnetic Bethe lattice • $\beta t = 25$ • $t = 0.25\,\mathrm{eV}$ • $T = 116.05\,\mathrm{K}$ • CT-HYB / DMFT



No smoothing; Hartree term $\Sigma_\infty = Un/2$ removed from real part. Shading = ±1 replica SEM.

Interacting Green function on the complete imaginary-time grid

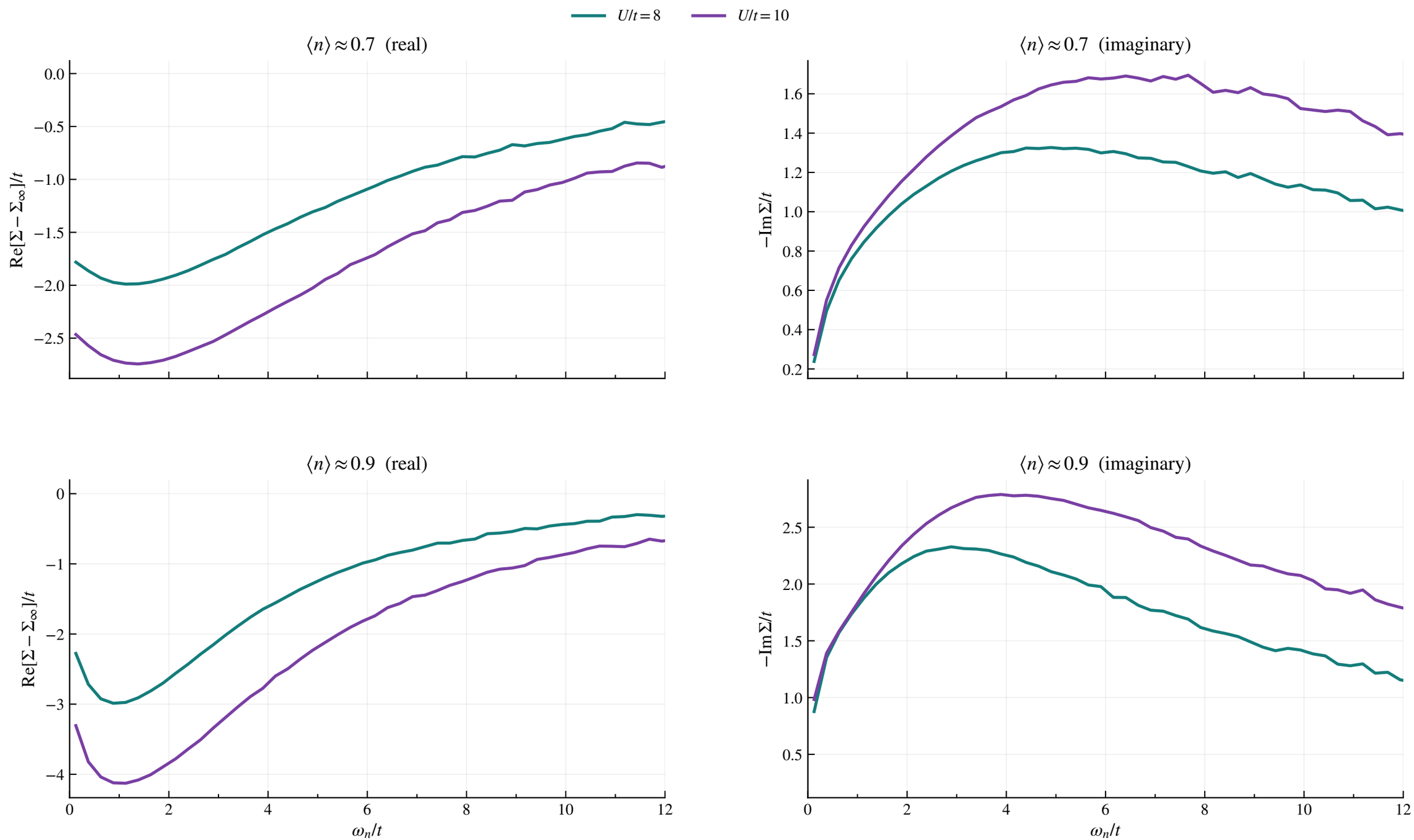
Paramagnetic Bethe lattice • $\beta t = 25$ • $t = 0.25$ eV • $T = 116.05$ K • CT-HYB / DMFT



Direct CT-HYB $G(\tau)$ (2049 grid points); measured endpoint density differs from the density-matrix estimate.

Self-energy: effect of interaction at fixed filling

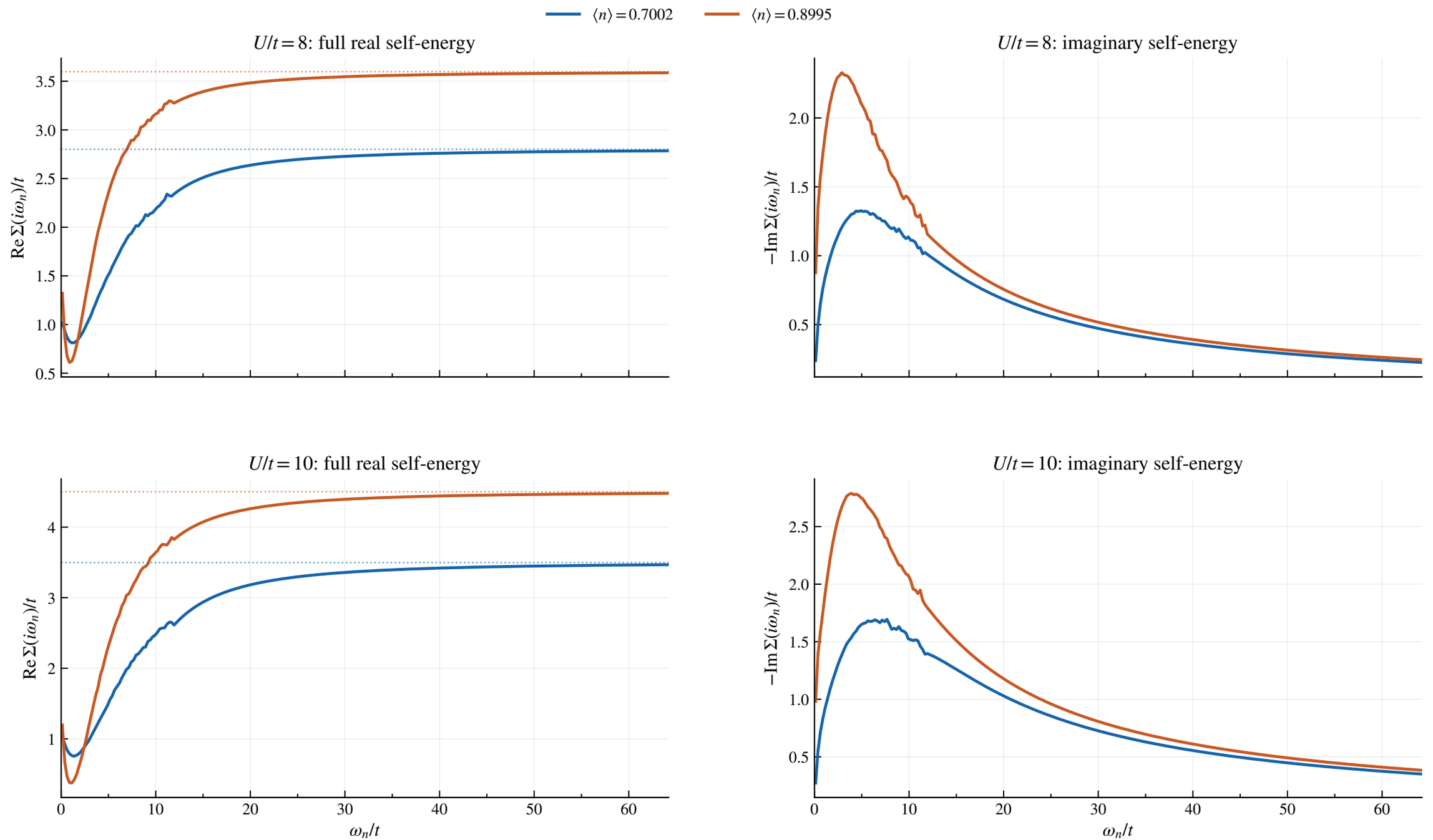
Paramagnetic Bethe lattice • $\beta t = 25$ • $t = 0.25\,\mathrm{eV}$ • $T = 116.05\,\mathrm{K}$ • CT-HYB / DMFT



For a fixed filling, compare $U/t = 8$ and 10 directly; Hartree shift removed from real part.

Complete self-energy (Hartree contribution retained)

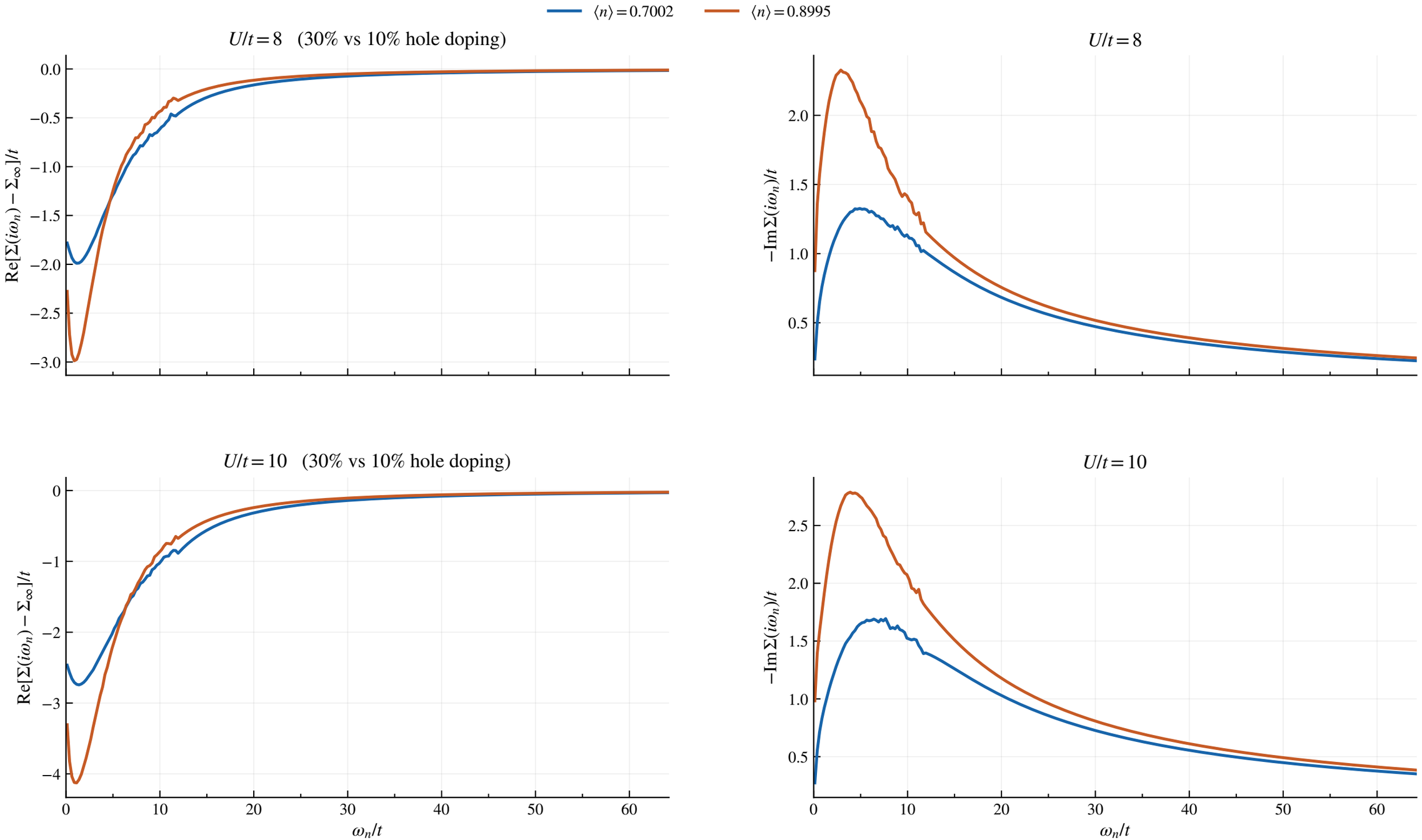
Paramagnetic Bethe lattice • $\beta t = 25$ • $t = 0.25\,\mathrm{eV}$ • $T = 116.05\,\mathrm{K}$ • CT-HYB / DMFT



Dotted levels: $\Sigma_\infty = Un/2$; all 256 positive Matsubara frequencies; no smoothing.

Self-energy: effect of doping at fixed interaction

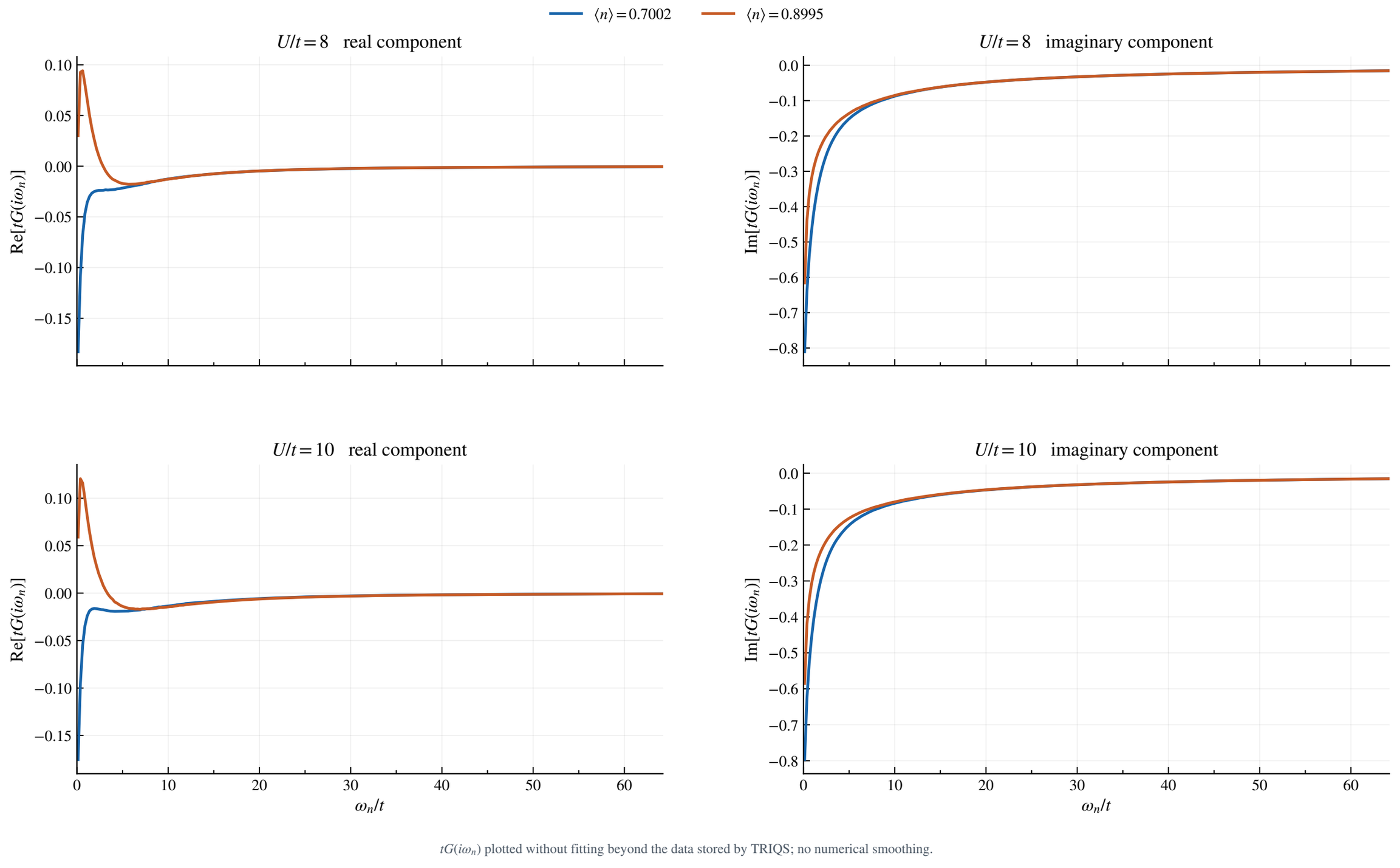
Paramagnetic Bethe lattice • $\beta t = 25$ • $t = 0.25\,\mathrm{eV}$ • $T = 116.05\,\mathrm{K}$ • CT-HYB / DMFT



Full positive Matsubara grid; tail fitted above $\omega_n/t \approx 12$.

Local interacting Green function: comparison of four cases

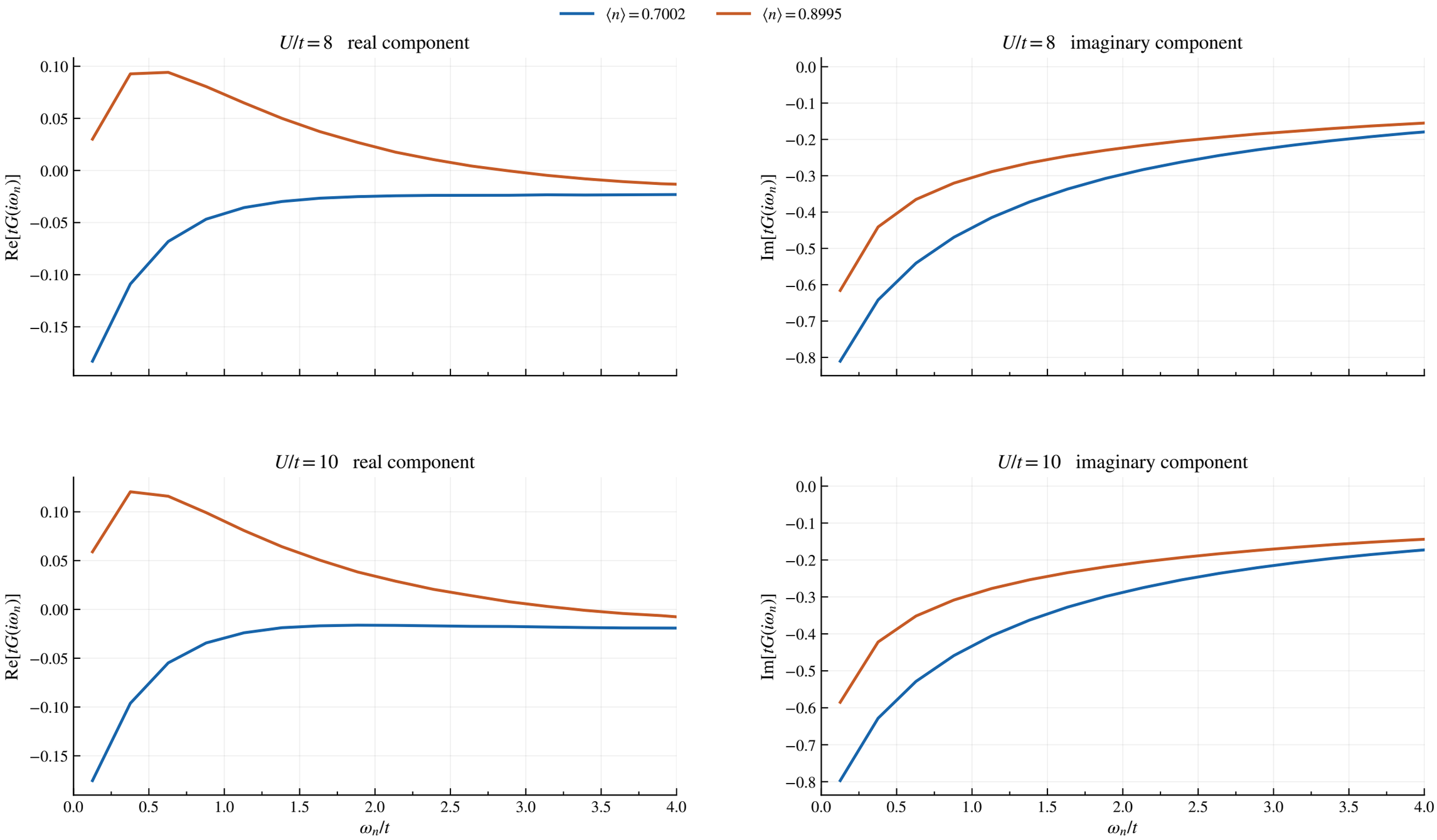
Paramagnetic Bethe lattice • $\beta t = 25$ • $t = 0.25\,\mathrm{eV}$ • $T = 116.05\,\mathrm{K}$ • CT-HYB / DMFT



$tG(i\omega_n)$ plotted without fitting beyond the data stored by TRIQS; no numerical smoothing.

Local interacting Green function: comparison of four cases

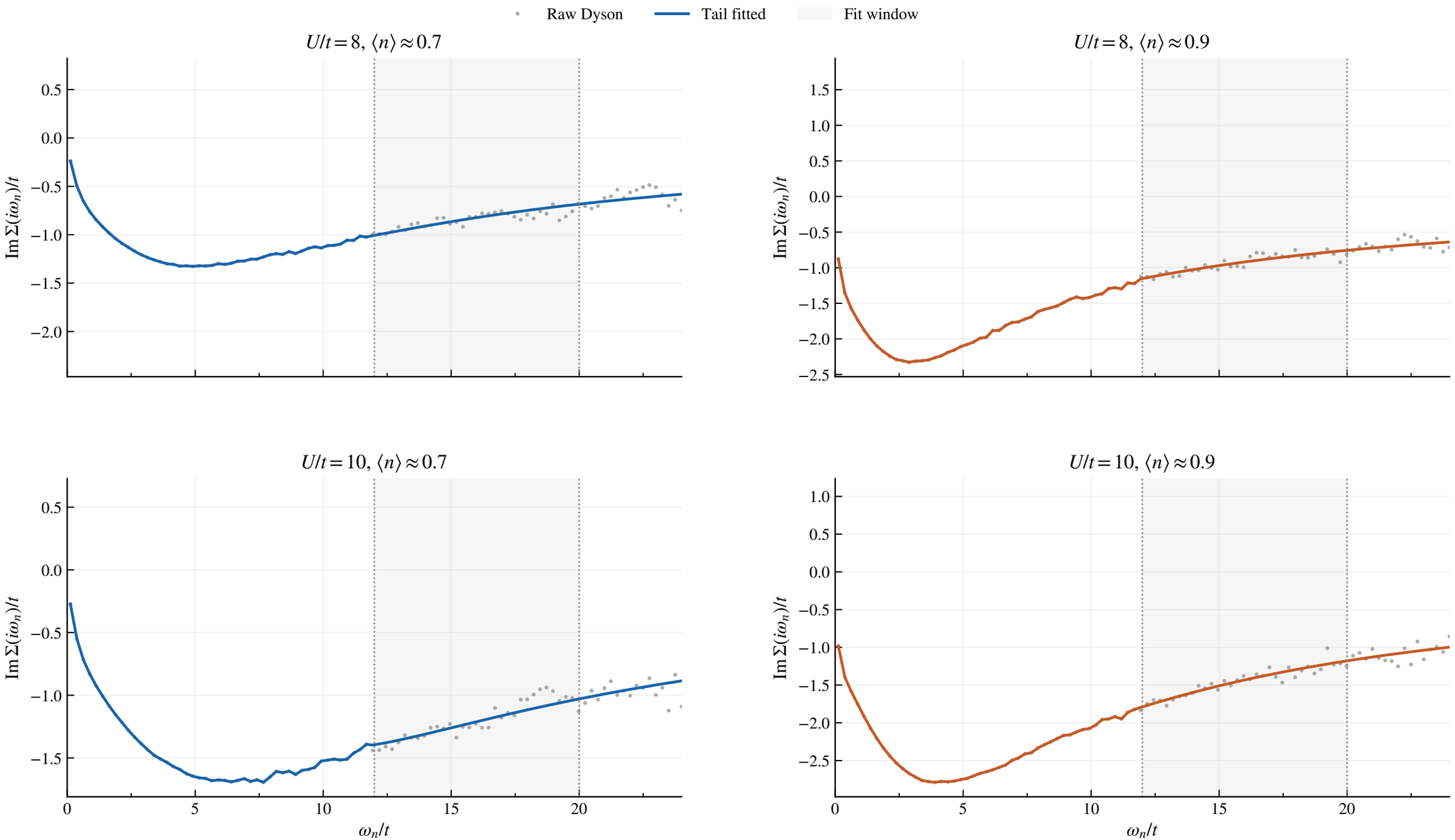
Paramagnetic Bethe lattice • $\beta t = 25$ • $t = 0.25\,\mathrm{eV}$ • $T = 116.05\,\mathrm{K}$ • CT-HYB / DMFT



$tG(i\omega_n)$ plotted without fitting beyond the data stored by TRIQS; no numerical smoothing.

Tail-fit diagnostic: raw versus processed self-energy

Paramagnetic Bethe lattice • $\beta t = 25$ • $t = 0.25\,\mathrm{eV}$ • $T = 116.05\,\mathrm{K}$ • CT-HYB / DMFT

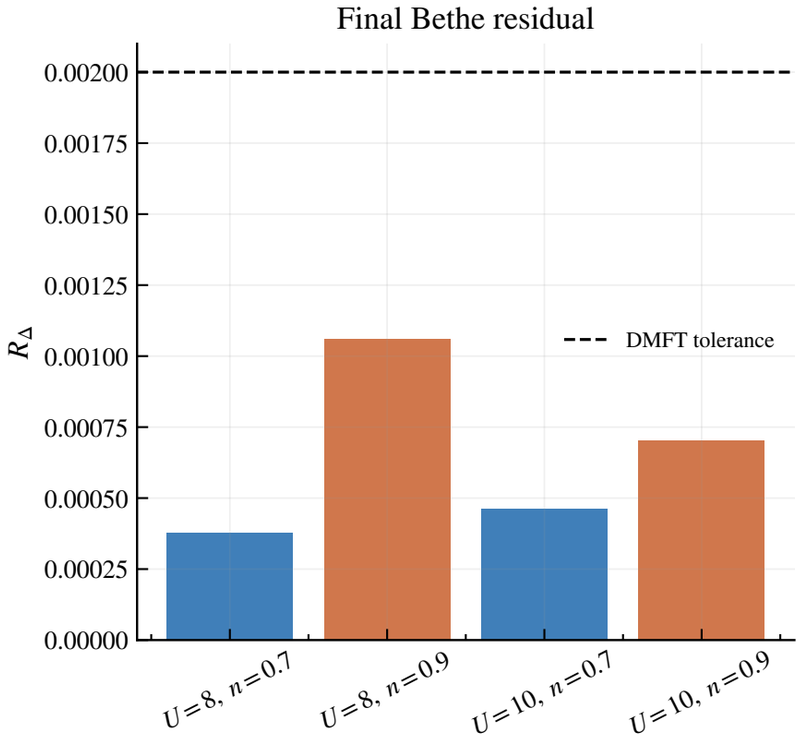


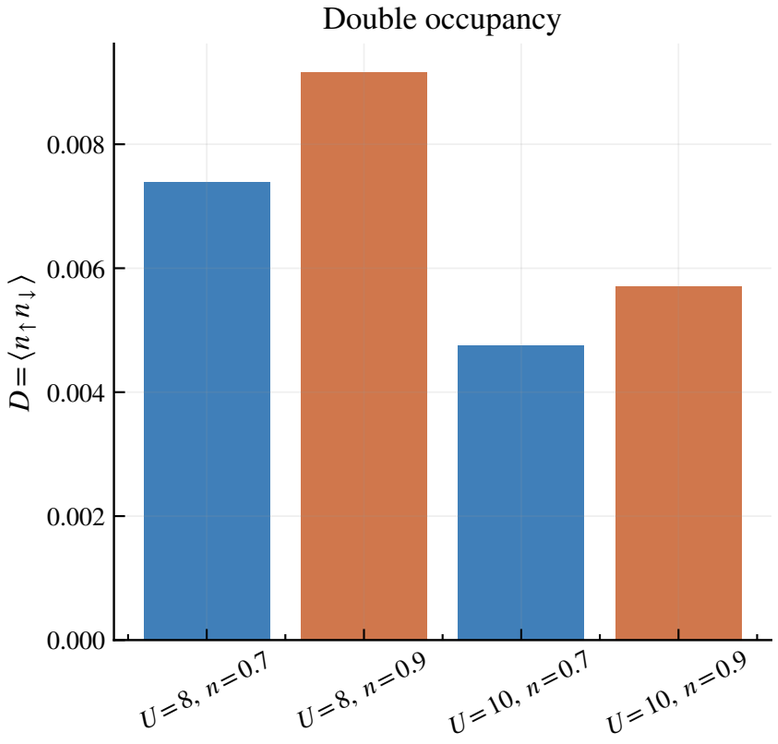
Raw QMC/Dyson estimates (points) and tail-fitted curves; do not interpret high-frequency raw noise as physics.

Numerical quality and correlation diagnostics

Paramagnetic Bethe lattice • $\beta t = 25$ • $t = 0.25\,\mathrm{eV}$ • $T = 116.05\,\mathrm{K}$ • CT-HYB / DMFT



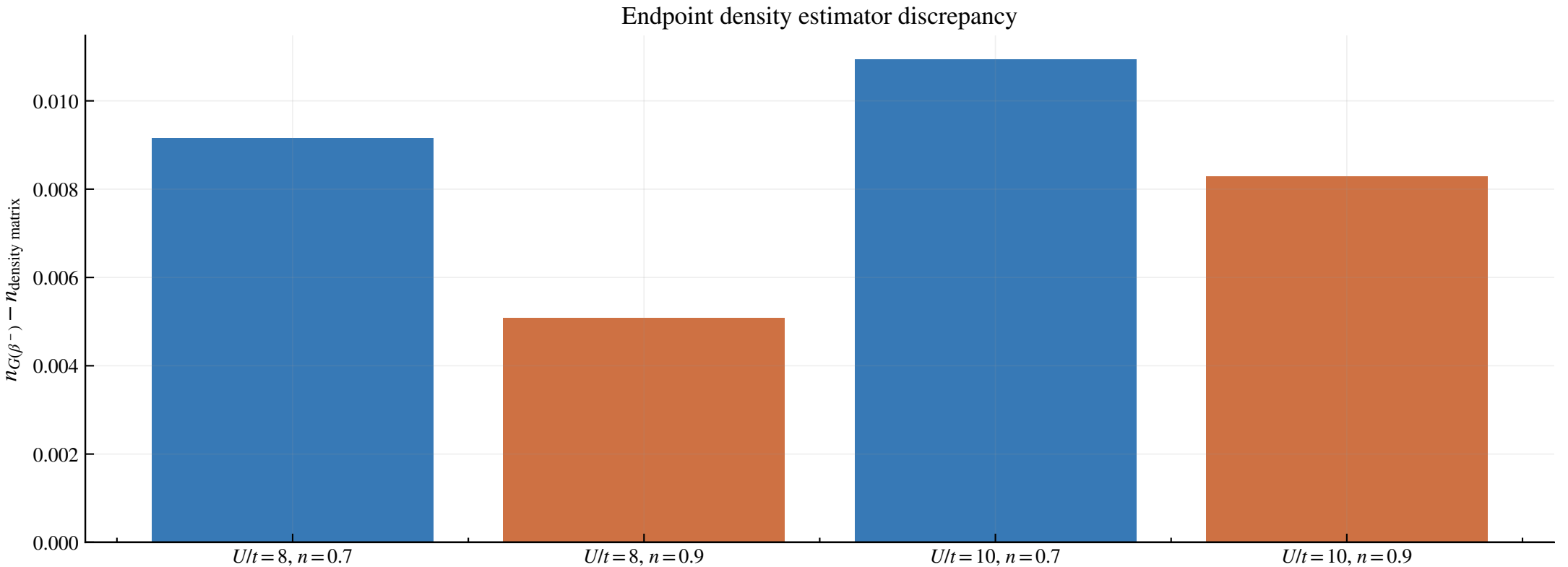




All four cases pass programmed tolerances; replica SEM excludes systematic bath and chemical-potential uncertainty.

Diagnostic: direct G(tau) endpoint density cross-check

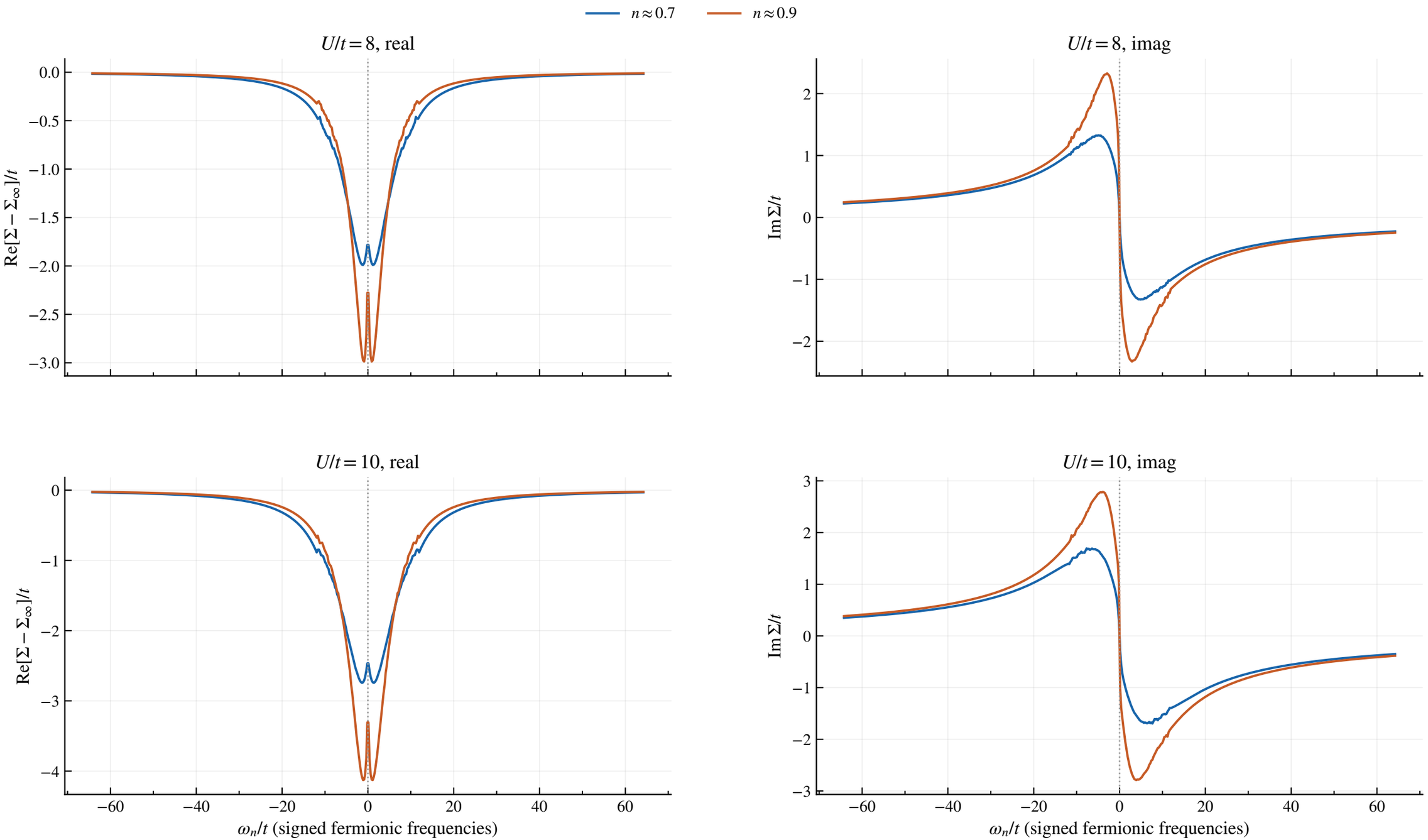
Paramagnetic Bethe lattice • $\beta t = 25$ • $t = 0.25$ eV • $T = 116.05$ K • CT-HYB / DMFT



This is an unresolved estimator/reconstruction discrepancy, not evidence that the Matsubara density is incorrect.

Self-energy over the full signed 512-point Matsubara mesh

Paramagnetic Bethe lattice • $\beta t = 25$ • $t = 0.25\,\mathrm{eV}$ • $T = 116.05\,\mathrm{K}$ • CT-HYB / DMFT



Negative-frequency data reconstructed via $\Sigma(-i\omega_n) = \Sigma(i\omega_n)^*$; not additional independent QMC samples.